Japan: Lessons for India

- Separate systems for large home appliances and small electronics clarify responsibilities.

- Retailers, manufacturers, municipalities, and certified recyclers have distinct, enforced roles.

- 14.95 million specified appliances received by manufacturers in FY 2022 (Japan).

- India lesson: make device return easy and track each handover across the chain.

Source: Case study report — Japan appliance recovery data, FY 2022.

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Switzerland’s E-Waste Success

- Take-back supported by dense collection points, retail returns, and centralized recycling finance (SENS).

- SENS reported recycling 100,762 tonnes in 2024 through its managed system (not entire national total).

- 442 official SENS collection points recorded in 2024, increasing convenience for consumers.

- India lesson: pair convenient returns with independently verified treatment.

Source: SENS system reports, 2024. Note: SENS ≠ full Swiss e-waste total.

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India’s E-Waste Management Rules 2022

- Rules came into force 1 April 2023, establishing registration and mandatory EPR mechanisms.

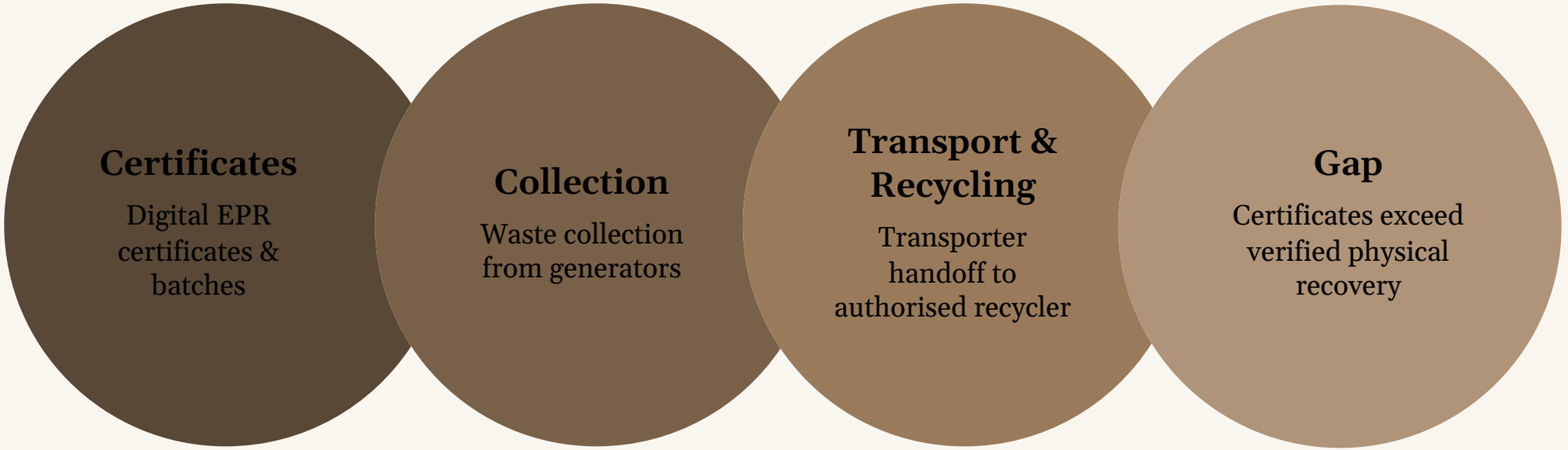
- EPR requires producers to register, submit targets, and arrange authorised collection and treatment.

- Estimated generation: 1,397,955.59 MT in FY 2024–25 (India, national estimate).

- Combined reported share collected, dismantled, and recycled/disposed: 70.71% (reporting metric; not pure material recycling rate).

Source: India case study report and national EPR filings, FY 2024–25. Figures include scope and year on slide.

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India’s EPR: Results or Paper Compliance?

- Producers fulfil obligations via registration and transferable certificates tied to waste volumes.

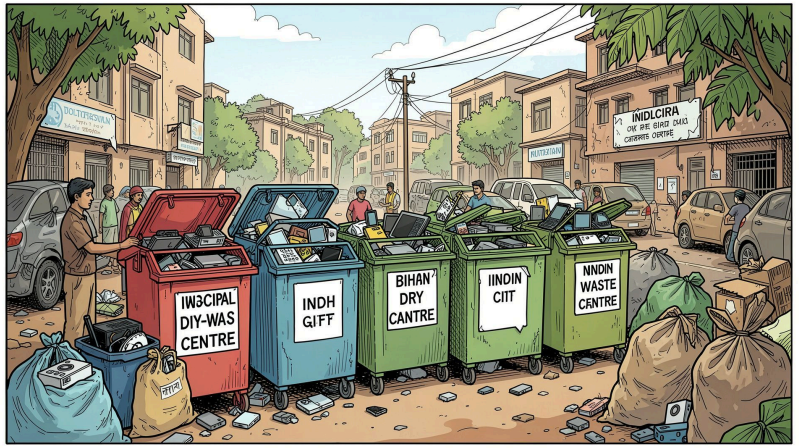
- Certificates must correspond to actual waste received and processed by authorised facilities.

- Main risk: administrative certificates can appear compliant even when physical recovery is inadequate.

- Solution: batch tracking, material-balance audits, and independent third-party verification.

Source: Case study report — EPR implementation observations (India, 2023–25).

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Bengaluru’s E-Waste Challenge

- High electronics use necessitates accessible local collection and clear routes to authorised recyclers.

- Municipal dry-waste centres present an opportunity to add separate e-waste intake streams.

- Karnataka registered recyclers processed 16,151.766 MT in FY 2024–25 (statewide figure).

- Priority: publish Bengaluru-specific generation and collection figures to guide local policy.

Source: Karnataka state recycler reports, FY 2024–25. Karnataka total, not Bengaluru-only.

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Mumbai’s E-Waste Ecosystem

- Multiple sources: households, offices, repair shops, institutions — each needs routing to authorised channels.

- Ward-level collection and sorting must link directly to registered recyclers to preserve traceability.

- Maharashtra registered recyclers processed 52,597.803 MT in FY 2024–25 (statewide figure).

- Priority: document every transfer and publish Mumbai-level performance data.

Source: Maharashtra recycler returns, FY 2024–25. Statewide number; slide shows year and scope.

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Kerala’s Clean Kerala Company Model

- Haritha Karma Sena supports household collection and segregation at source, improving material quality.

- Local facilities and Clean Kerala Company coordinate collection, aggregation, and links to recyclers.

- Kerala registered e-waste recyclers processed 10,079.632 MT in FY 2024–25 (statewide figure).

- Key action: track e-waste separately from general inorganic waste streams for measurement and control.

Source: Kerala state recycling reports, FY 2024–25. State total; not exclusively Clean Kerala Company output.

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Pune’s SWaCH Model

- SWaCH: a waste-picker owned cooperative working with Pune Municipal Corporation to formalize collection services.

- Authorised for e-waste collection since 2012; uses collection bins and drives to reach households.

- System design: route collected devices to authorised recycling partners to maintain traceability and safety.

- Lesson: protect livelihoods while improving safety, formal recognition, and material traceability.

Source: SWaCH partnership documentation and pilot reports. Note: early pilot worker/household figures are not e-waste tonnages.

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Dharavi’s Recycling Ecosystem

- Repair, resale, collection, and informal material recovery underpin local livelihoods and circular flows.

- Key concerns: unsafe dismantling practices and untracked hazardous residues harming workers and environment.

- Support needed: training, PPE, formal links to authorised recyclers, and pathways for safe residue management.

- Research priority: conduct a dedicated survey before reporting any Dharavi-level tonnage or recycling rate.

Source: Field observations and the case study report. Emphasize need for measured local survey before tonnage claims.

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Dell’s Circular Economy Strategy

- Design for repair and disassembly; corporate take-back enables reuse and recycling of components.

- Dell target for 2030: reuse or recycle one tonne for every tonne sold (company target, not yet achieved).

- Example: Latitude 9440 uses recycled materials in selected components to lower virgin material demand.

- India lesson: pair statutory producer compliance with product longevity, repairability, and clear take-back channels.

Closing line: Effective e-waste management connects collection, worker safety, verified recycling, and circular product design.

Source: Dell corporate sustainability targets and product disclosures. 2030 goal presented as target (company stated).

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